Original Query (Q)

```
1: SELECT COUNT(*)
2: FROM PostHistory ph,
3:   Posts p,
4:   Users u,
5:   Badges b
6: WHERE b.UserId = u.Id
7:   AND p.OwnerUserId = u.Id
8:   AND ph.UserId = u.Id
9:   AND ph.CreationDate >= '2010-07-27 18:08:19'
10:  AND ph.CreationDate <= '2014-09-10 08:22:43'
11:  AND p.PostTypeId = 2;
```

Rewrite Engine

Fragment Index (FI)

$\mathcal{K}_1$: AGG(FILTER(JOIN[CROSS]))
$\mathcal{K}_2$: FILTER(JOIN[CROSS])
$\mathcal{K}_3$: FILTER

FI lookup: 18 rules, 2 fire

R_1: AggregateDecomposeCrossJoin
R_2: PredicateRangeToBetween

Rule Order $\Rightarrow \{R_1 \rightarrow R_2\}$

Rewritten Query (Q')

```
1: WITH ph_counts AS (SELECT UserId, COUNT(*) AS ct
2:   FROM PostHistory WHERE CreationDate
3:   BETWEEN '2010-07-27 18:08:19' AND '2014-09-10 08:22:43' GROUP BY UserId),
4: p_counts AS (SELECT OwnerUserId, COUNT(*) AS ct FROM Posts
5:   WHERE PostTypeId = 2 GROUP BY OwnerUserId),
6: u_counts AS (SELECT Id, COUNT(*) AS ct FROM Users GROUP BY Id),
7: b_counts AS (SELECT UserId, COUNT(*) AS ct FROM Badges GROUP BY UserId)
8: SELECT COALESCE(SUM(ph.ct * p.ct * b.ct * u.ct), 0)
9: FROM u_counts AS u
10:  INNER JOIN ph_counts AS ph ON u.Id = ph.UserId
11:  INNER JOIN p_counts  AS p  ON u.Id = p.OwnerUserId
12:  INNER JOIN b_counts  AS b  ON u.Id = b.UserId;
```